



## Original Contribution

# Ultrasound-guided bilateral serratus plane block versus thoracic paravertebral block following minimally invasive pectus excavatum repair in children: A randomized controlled non-inferiority study.

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## HIGHLIGHTS

- SAPB showed comparable pain control to TPVB for Nuss procedure analgesia in children.
- Ultrasound-guided SAPB is a viable alternative for pediatric thoracic analgesia.
- SAPB had fewer adverse effects, suggesting better safety and tolerance.

## ARTICLE INFO

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## ABSTRACT

**Study objective:** This study evaluated whether ultrasound-guided serratus anterior plane block (SAPB) provided non-inferior analgesic effects for minimally invasive pectus excavatum repair surgery compared with thoracic paravertebral block (TPVB).

**Design:** A noninferiority randomized trial.

**Setting:** West China Hospital of Sichuan University.

**Patients:** Seventy-four children aged 7–16 years who underwent minimally invasive pectus excavatum repair surgery were enrolled.

**Intervention:** Patients were randomly assigned to receive bilateral SAPB ( $n = 37$ ) or TPVB ( $n = 37$ ) after induction of anesthesia ( $0.5 \text{ mL kg}^{-1}$  0.25 % of ropivacaine per side).

**Measurements:** Pain scores assessed via numerical rating scale (NRS) postoperatively; opioid consumption, block-related complications; and plasma ropivacaine concentrations were measured.

**Main results:** Median (IQR) pain scores for SAPB were 1 (1.0–2.0) and 1 (0.5–2.0) for those with TPVB 24 h postoperatively (effect size = 0.027; 95 % confidence interval,  $-0.42$  to  $0.47$ ,  $P = 0.905$ ), meeting the non-inferiority criterion with a pre-specified margin of 0.5. The TPVB group exhibited a greater incidence of hypotension (29.7 % vs. 8.1 %,  $p = 0.018$ ). The ropivacaine concentrations were lower in the SAPB group at all measured time points (SAPB: 0.44 (0.21), 0.56 (0.23), and 0.66 (0.29)  $\mu\text{g mL}^{-1}$  vs. TPVB: 1.18 (0.39), 1.17 (0.30), and 1.13 (0.26)  $\mu\text{g mL}^{-1}$  at 10, 30, and 60 min post-injection, respectively).

**Conclusion:** Compared with TPVB, SAPB provides non-inferior analgesia for children undergoing minimally invasive pectus excavatum repair surgery. Moreover, SAPB is associated with less intraoperative hemodynamic instability and lower plasma concentrations, suggesting it is a safe and valid alternative.

**Trial registration:** Chinese Clinical Trial Registry, identifier: ChiCTR 2,200,056,596.

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## 1. Introduction

The current first-line approach for pediatric pectus excavatum repair is minimally invasive repair of pectus excavatum (MIRPE), commonly known as the Nuss procedure [1]. This technique involves placing stainless steel or titanium bars beneath the sternum under thoracoscopic guidance, eliminating the need for cartilage resection or osteotomy [1,2]. Although this minimally invasive approach reduces direct tissue trauma, the placement of curved metal bars behind the sternum—an innovation that has revolutionized chest wall reconstruction—is associated with significant postoperative pain [2–4]. Severe immediate postoperative pain may contribute to the development of persistent psychological symptoms and chronic pain, potentially impacting the long-term quality of life in pediatric patients [5]. Consequently, effective perioperative pain management is a crucial aspect of the hospital course.

Traditional opioid-based analgesia in children is often administered intermittently as a single agent, which can be effective but carries risks such as respiratory depression and nausea [6]. Owing to the ongoing opioid crisis, the current pain management paradigm has shifted toward multimodal analgesia incorporating regional and neuraxial techniques. Optimizing postoperative chest wall pain management continues to be a key focus in anesthesiology.

Historically, thoracic epidural analgesia (TEA) has long been the standard of care for managing thoracotomy pain. However, TEA requires specialized expertise and is associated with potential complications [7]. Thoracic paravertebral block (TPVB) is recognized as an effective alternative to TEA, but its routine use in pediatric patients is limited due to its technical complexity, procedural time, and risks such as pleural injury and hemodynamic instability [8]. With the advancement of ultrasound-guided techniques, peripheral nerve blocks have gained popularity as safer and more feasible alternatives [9]. Among them, the serratus anterior plane block (SAPB) is an emerging, low-risk interfascial plane block that targets the lateral cutaneous branch of the intercostal nerves, as well as the long thoracic nerve and thoracodorsal nerves. SAPB produces dermatomal anesthesia from T2 to T9, showing promise for pain relief and opioid reduction in patients undergoing the Nuss procedure. Recent studies have specifically demonstrated that ultrasound-guided serratus anterior plane block (SAPB) can effectively reduce postoperative pain and opioid requirements in pediatric and adolescent patients undergoing MIRPE [10]. Although some evidence drawn from thoracic surgery suggests that SAPB may provide analgesia comparable to thoracic paravertebral block (TPVB) when used as part of multimodal analgesia regimens [11–13], other data indicated that single-shot SAPB could be associated with greater postoperative opioid requirements compared with TPVB [14]. However, direct comparative studies focusing on SAPB versus TPVB specifically in the context of MIRPE remain scarce, leading to uncertainty about their relative effectiveness for this unique procedure, which differs from thoracotomy and VATS in both surgical approach and pain mechanisms. On the basis of these considerations, we hypothesized that, in pediatric patients undergoing the Nuss procedure, single-shot ultrasound-guided bilateral SAPB would be non-inferior to bilateral TPVB in terms of effective pain management.

## 2. Materials and methods

### 2.1. Ethics and registration

This study received ethical approval from the Institutional Ethics Committee of West China Hospital, Sichuan University, China (approval number 20211275), approved by Chairperson Prof. Shaolin Deng, on 11 November 2021, following the original protocol which was previously reported [15]. Written informed consent was obtained from all patients or their legal guardians after a detailed explanation of the planned intervention. The trial was prospectively registered at the Chinese

Clinical Trial Registry (ChiCTR 2,200,056,596, investigator, Mingzhe Xu; date of registration, 08 February 2022, accessible at [chiCTR.org.cn/index.aspx](http://chiCTR.org.cn/index.aspx)) before the commencement of patient enrollment. The study adhered to the Consolidated Standards of Reporting Trials (CONSORT) guidelines.

### 2.2. Patients' inclusion and exclusion criteria

We recruited patients aged 7 to 16 years who were scheduled for Nuss surgery and an American Society of Anesthesiologists (ASA) physical status of I to III. The exclusion criteria included patients requiring prolonged intubation for more than 12 h, those with infection or anatomical abnormalities at the injection site, those with a history of previous Nuss procedures, those with known allergies to ropivacaine, and those who declined to provide informed consent.

### 2.3. Randomization and masking

Patients were randomly assigned to the SAPB or TPVB group with a 1:1 allocation ratio via a computer-generated number created by an independent statistician (MZX). Each patient's code and group allocation were enclosed in a sealed envelope. Regional blocks were administered by an experienced anesthesiologist (DB) who was not involved in intraoperative management or data collection. All surgeries were conducted by a fixed surgical team. Postoperative follow-up was recorded in case report forms by trained researchers (ZL and YJQ) who were not involved in nerve block administration or intraoperative management. The allocation was blinded to the anesthesiologists (responsible for intraoperative care and follow-up), surgeons, participants, and statisticians until the final analyses were completed.

### 2.4. Protocol

Induction, maintenance, and emergence from general anesthesia were performed according to the study protocol. [15] Standard intraoperative monitoring included 5-lead electrocardiography, pulse oximetry, capnography, radial artery blood pressure monitoring, and bispectral index (BIS) monitoring. Anesthesia was induced with propofol ( $2.5\text{--}3\text{ mg kg}^{-1}$ ), sufentanil ( $0.3\text{--}0.5\text{ }\mu\text{g kg}^{-1}$ ), and cisatracurium ( $0.1\text{--}0.2\text{ mg kg}^{-1}$ ). After intubation, patients in the SAPB group received a deep serratus anterior plane block at the third and fourth ribs, whereas those in the TPVB group received a thoracic paravertebral block at the fifth vertebral level after the prone position was changed, with a cumulative dosage of  $2.5\text{ mg kg}^{-1}$  0.25 % ropivacaine (with 1:200,000 epinephrine). All blocks were performed by a single experienced anesthesiologist (DB) via a high-frequency linear transducer (6–14 MHz, M9, Mindray BioMedical Electronics, Shenzhen, China) and a nerve block needle (B Braun Aesculap, Japan Co;  $0.71 \times 80\text{ mm}$ ). Anesthesia was maintained with 1.5 MAC sevoflurane to keep the bispectral index at 40–60, using a 50 % air/oxygen mixture and volume-controlled ventilation ( $6\text{--}8\text{ mg kg}^{-1}$ ). Sufentanil ( $0.1\text{ }\mu\text{g kg}^{-1}$  per dose) were intermittently administered to maintain hemodynamic parameters (heart rate or mean arterial pressure) within 20 % of the preoperative baseline values. All procedures were performed by two experienced pediatric anesthesiologists at our center, which helped ensure consistent clinical decision-making. After the procedure, patients received ondansetron, and neuromuscular blockade was reversed with neostigmine. A patient-controlled analgesia (PCA) device was connected 30 min before the end of surgery, containing sufentanil ( $2\text{ }\mu\text{g kg}^{-1}$ ) and ondansetron ( $0.1\text{ mg kg}^{-1}$ ). The pump was programmed to deliver a background infusion at  $2\text{ mL/h}$  and patient-controlled boluses of  $0.5\text{ mL}$  with a 15-min lockout. All patients were received ibuprofen ( $10\text{ mg kg}^{-1}$ ) every 8 h for pain management. If a top-up dose fails to provide adequate pain relief (Numerical Rating Scale (NRS) scores  $>4$ ),  $0.1\text{ }\mu\text{g kg}^{-1}$  sufentanil in the PACU or ibuprofen ( $10\text{ mg kg}^{-1}$ ) in the ward were used as rescue analgesics. Thoracic paravertebral block was performed as described in

the protocol after intubation, with the patient positioned prone [15]. Using an in-plane technique, a nerve block needle (B Braun Aesculap, Japan Co;  $0.71 \times 80$  mm) was inserted, advancing from a lateral to medial direction, under ultrasound guidance at the paravertebral space at the fifth vertebral level. After the superior costotransverse ligament was perforated and a good view of the needle tip was confirmed, 0.5–1 mL of normal saline was injected to observe the anterior movement of the pleura. Subsequently, 0.25 % ropivacaine with 1:200,000 epinephrine, at a dose of  $0.5 \text{ mL kg}^{-1}$ , was administered into the unilateral paravertebral space. The same technique was used to perform the paravertebral block on the contralateral side.

Serratus anterior plane block was performed as outlined in the protocol using the ultrasound-guided technique described by Blanco et al. [16] With the patient in the supine position and the upper arm resting slightly over the head, and under direct visualization of the ultrasound probe, a nerve block needle (B Braun Aesculap, Japan Co;  $0.71 \times 80$  mm) was inserted into the plane between the serratus anterior and latissimus dorsi muscles along the midaxillary line of the third and fourth ribs, following preparation of the block area. Once adequate visualization of the needle tip was achieved, 0.5 mL of normal saline was injected to confirm plane separation via hydro-dissection. Subsequently, 0.25 % ropivacaine with 1:200,000 epinephrine, at a total dose of  $1 \text{ mL kg}^{-1}$ , was administered bilaterally into the fascial plane.

## 2.5. Measurements

As a primary outcome, the average pain score at rest 24 h after surgery was measured via the National Pain Rating Scale (NRS), which categorizes pain severity into no pain (0), mild pain (1–3), moderate pain (4–6), and severe pain (7). The secondary outcomes included the time required to perform the bilateral block, intraoperative opioid consumption, consumption of analgesic medication within 24 h postoperatively at morphine-equivalent doses (<https://globalrph.com/>), time to first analgesic request, total number of PCA presses (both effective and ineffective), pain scores during movement assessed at 1, 3, 6, and 24 h postsurgery, extubation time, and the incidence of nonsurgical complications (postoperative nausea and vomiting, hypotension, urinary retention, or local anesthetic toxicity). Ropivacaine concentrations were measured at 10, 30, and 60 min after the regional blocks were removed via liquid chromatography-tandem mass spectrometry.

## 2.6. Sample size calculation

The sample size was calculated via PASS software V.15.0.5 (PASS, LLC, Kaysville, UT, USA). The results of our previous study and preliminary experimental data were used to estimate the required sample size, in which the average pain score at 24 h after TPVB was 2.58 (0.92). We set the non-inferiority margin at 0.5, and group sample sizes of 35 per group provided 90 % power to detect non-inferiority via a one-sided, two-sample *t*-test. The significance level ( $\alpha$ ) of the test was 0.025. Considering an overall withdrawal rate of 5 %, the final sample size was increased to 37 patients per group.

## 2.7. Statistical analysis

All analyses were conducted on the basis of the intention-to-treat (ITT) principle. Statistical calculations were performed via SPSS software (version 23.0, IBM SPSS Inc., Chicago, IL, USA). The Kolmogorov–Smirnov test was applied to assess the normality of continuous outcomes. For normally distributed continuous data, results are presented as the means (standard deviations, SDs), whereas non-normally distributed continuous data are reported as medians with interquartile ranges (IQRs). Normally distributed continuous data were compared via the unpaired Student's *t*-test, whereas nonparametric data were analyzed via the Mann–Whitney *U* test. Categorical data were compared via the chi-square test. Intergroup comparisons of pain scores were

analyzed via repeated-measures ANOVA, followed by post hoc pairwise comparisons or Bonferroni adjustment. Statistical significance was considered at  $P < 0.05$ .

## 3. Results

From June 2022 to February 2024, a total of ninety-one children scheduled for Nuss procedures were enrolled in the study (Fig. 1). After enrollment, six patients opted to withdraw before surgery and randomization, six patients had their surgeries cancelled before randomization, and five patients could not be randomized due to disruptions in research workflows during the COVID-19 pandemic (Fig. 1). The remaining seventy-four participants were assigned to two groups, TPVB ( $n = 37$ ) and SAPB ( $n = 37$ ), according to a previously generated random allocation. All randomized patients completed follow-up, and their data were included in the final analysis. All blocks were successfully completed without any noticeable complications, such as bleeding at the puncture site or intravascular injection. Demographics, baseline assessments, and intraoperative details were similar between the 2 groups and are summarized in Table 1. There were no significant differences between the TPVB and SAPB groups in terms of patient characteristics (age, sex and BMI), duration of anesthesia or surgery, or extubation time.

### 3.1. Primary outcome

The median NRS pain scores at rest or during movement were comparable across all timepoints in both groups (Fig. 2). For the primary outcome, the median (IQR) pain score at rest 24 h after surgery for participants with SAPB was 1 (1.0–2.0) vs 1 (0.5–2.0) for those with TPVB (effect size = 0.027; 95 % confidence interval [CI],  $-0.42$  to  $0.47$ ,  $P = 0.905$ ). Given that the prespecified framework margin was set at 0.5, non-inferiority was demonstrated. On the basis of the composite of all available pain measures, patients receiving TPVB demonstrated lower composite pain scores than did those receiving SAPB during the initial 1-h postoperative period at rest, although the difference did not reach statistical significance (Fig. 2).

### 3.2. Secondary points

Intraoperative sufentanil administration was greater in the SAPB group than in the TPVB group ( $0.54$  [ $0.96$ ]  $\mu\text{g kg}^{-1}$  vs.  $0.48$  [ $0.1$ ]  $\mu\text{g kg}^{-1}$ ,  $P = 0.005$ ; Table 2). However, postoperative opioid consumption within the first 24 h after extubation was comparable between the two groups ( $1.12$  [ $1.06$ ,  $1.21$ ]  $\text{mg kg}^{-1}$  morphine equivalents for the SAPB group vs  $1.11$  [ $1.07$ ,  $1.20$ ]  $\text{mg kg}^{-1}$  for the TPVB group,  $p = 0.863$ ; Table 2). Additionally, the time to the first analgesic request was similar for both groups ( $74$  [ $42.3173.5$ ] minutes vs.  $84$  [ $62.5155.5$ ] minutes,  $P = 0.559$ ; Table 2). The number of effective versus ineffective patient-controlled analgesia (PCA) presses also did not differ significantly between the two groups (Fig. 3). The median (IQR) duration required to perform bilateral SAPB was notably shorter than that required for TPVB, indicating that SAPB was a more time-efficient procedure ( $6$  [ $4.5, 9$ ] minutes vs.  $13$  [ $9, 15$ ] minutes,  $P = 0.000$ ; Table 1).

Hypotension was observed in three patients (8.1 %) in the SAPB group, and in eleven patients (29.7 %) in the TPVB group ( $p = 0.018$ ). Within 24 h postoperatively, four patients (10.8 %) in the SAPB group required urinary catheterization for retention, a complication not observed in the TPVB group. The incidence of postoperative nausea and vomiting (PONV) was similar between the groups, with nine patients (24.3 %) in the SAPB group and eight patients (21.6 %) in the TPVB group experiencing PONV ( $p = 0.782$ ). Additionally, oral analgesic rescue medication was administered to seven patients (18.9 %) in the SAPB group and to five patients (13.5 %) in the TPVB group ( $p = 0.528$ ).

With respect to the ropivacaine arterial plasma concentration, in the SAPB group, the average concentrations at 10, 30, and 60 min after

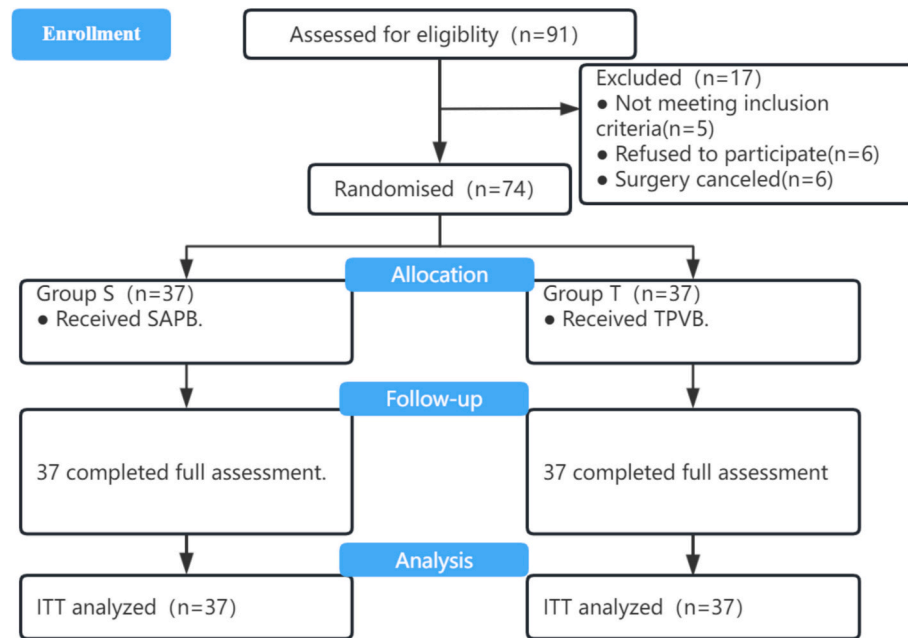


Fig. 1. CONSORT flow diagram.

Group T: thoracic paravertebral block group; Group S: serratus anterior plane block; ITT: intent to treat.

Table 1

Demographic and surgical details receiving TPVB or SAPB of Nuss procedures. Values are mean  $\pm$  SD or median (IQR [range]).

|                                | SAPB group<br>n = 37 | TPVB group<br>n = 37 | P Value |
|--------------------------------|----------------------|----------------------|---------|
| Age, years                     | 13(10.5,14)          | 12(10,13)            | 0.243   |
| Sex(male/female)               | 34/3                 | 28/9                 | 0.06    |
| BMI (kg m <sup>-2</sup> )      | 16.67 (15.90, 17.90) | 16.23(14.87, 17.44)  | 0.216   |
| Anesthesia duration, min       | 115(95,141)          | 116(95,139)          | 0.774   |
| Surgery duration, min          | 50(38.5,60.5)        | 50(35.5,66)          | 0.837   |
| Extubation time, min           | 6(2,12.5)            | 5(3,8.5)             | 0.66    |
| Time of regional blockage, min | 6(4.5,9)             | 13(9,15)             | 0.000*  |
| Extubation time, min           | 6(2,12.5)            | 5(3,8.5)             | 0.66    |
| hospital costs, thousand yuan  | 32.08 (31.27,33.73)  | 31.96 (31.44,34.31)  | 0.858   |
| Length of hospital stay, days  | 7(7,8)               | 7(7,8)               | 0.693   |

BMI: body mass index; TPVB: thoracic paravertebral block; SAPB: serratus anterior plane block.

\* Compared to control group,  $P < 0.05$ .

injection were  $0.44 [0.21] \mu\text{g mL}^{-1}$ ,  $0.56[0.23] \mu\text{g mL}^{-1}$ , and  $0.66[0.29] \mu\text{g mL}^{-1}$ , respectively. In the TVPB group, the mean plasma ropivacaine concentrations were  $1.18 [0.39] \mu\text{g mL}^{-1}$ ,  $1.17 [0.30] \mu\text{g mL}^{-1}$ , and  $1.13 [0.26] \mu\text{g mL}^{-1}$  at 10, 30, and 60 min after injection, respectively (Fig. 4). Significant interindividual variability in arterial plasma ropivacaine concentrations was observed in both groups.

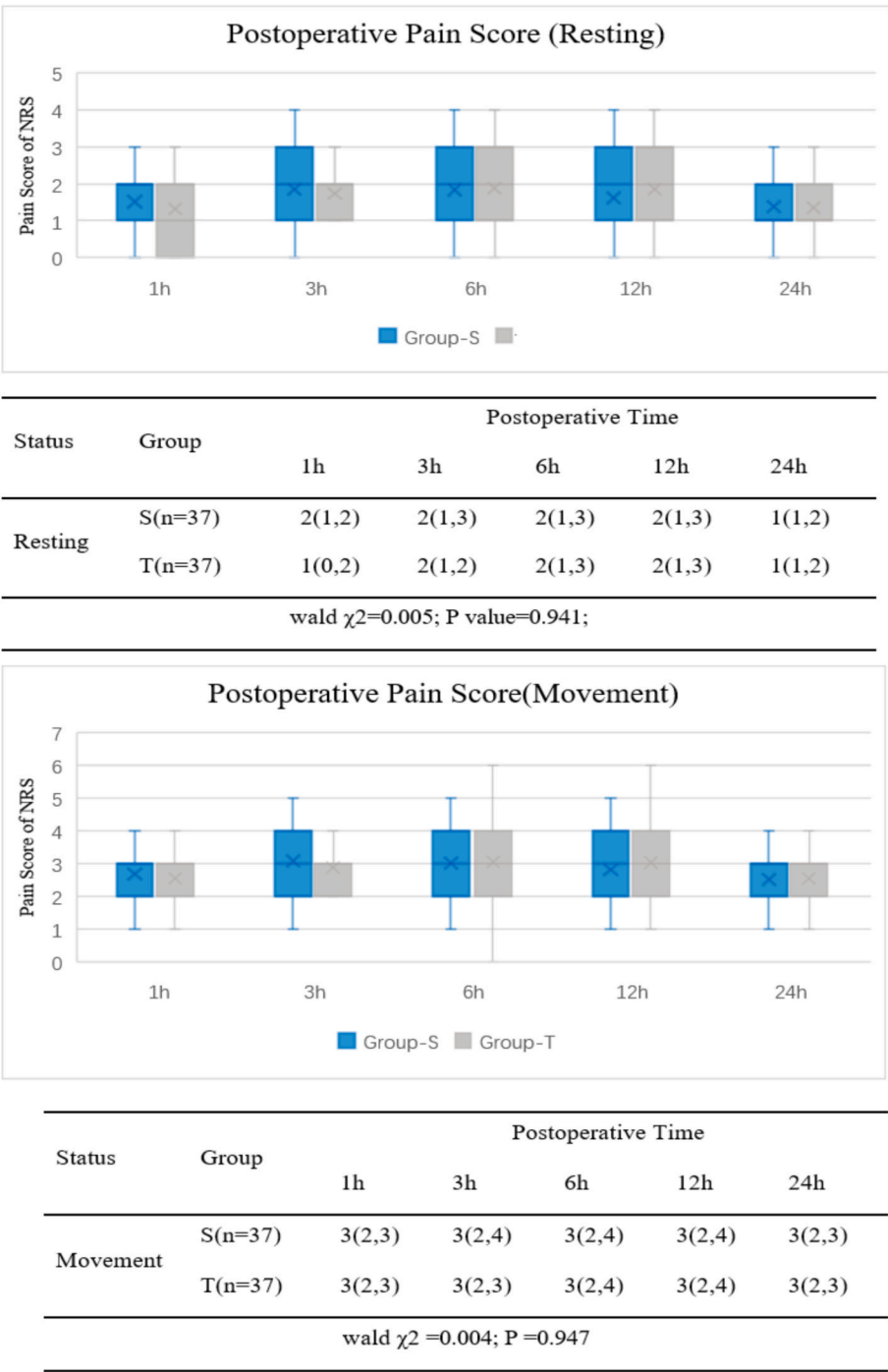
#### 4. Discussion

In this study, we found no significant differences in postoperative visual analogue scale (VAS) pain scores, either at rest or during coughing, between patients who received a serratus anterior plane block (SAPB) or a thoracic paravertebral block (TPVB) during the Nuss procedure in children. These findings confirm the non-inferiority of SAPB over TPVB in terms of postoperative analgesia. As nerve blocks are components of a multimodal analgesia regimen, the potential analgesic differences between the two block techniques may have been masked to

some extent by the other complementary analgesic strategies. In terms of opioid sparing during surgery, thoracic paravertebral blocks demonstrated superiority over serratus plane blocks. However, opioid consumption within the first 24 h postoperatively was comparable between the two groups. Several secondary outcomes favor SAPB, including a shorter procedural time and a lower incidence of intraoperative hypotension. Additionally, plasma ropivacaine concentrations were significantly lower in the SAPB group than in the TPVB group, suggesting a potentially improved safety profile.

The overall pain scores observed in our cohort were relatively low compared to some previously published studies on MIRPE. Similar findings have been reported by Altun et al., who observed mild median pain scores and reduced opioid consumption in patients receiving single-injection SAPB after MIRPE [10]. Previous studies have suggested that postoperative pain intensity after MIRPE tends to increase due to chest wall rigidity and higher bar pressure on the sternum and ribs [17]. Young children typically have more compliant chest walls, resulting in less bony resistance and potentially less nociceptive stimulation from substernal bar placement. Our patient cohort was younger, and the majority of patients in our study had only a single pectus bar implanted, whereas many previous studies involved patients requiring multiple bars due to more severe or complex deformities [12,13]. The discomfort may originate more from the forced expansion of the flexible chest wall rather than direct traction on stiffer costal cartilages or bones, which is more pronounced in older adolescents and adults [18,19].

Analgesic approaches to MIRPE are available nationwide [9]. In recent years, cryoablation (INC) has gained popularity in NUSS procedures at many centers and offers significant advantages [20,21]. However, its use remains somewhat controversial, considering its extra invasiveness, instrumentation and the potential risks such as chronic neuropathic pain [22,23]. Besides, cryoablation requires single-lung ventilation with double-lumen intubation, video-assisted thoracoscopic procedures, and specialized cryo-equipment, which together increase invasiveness, operative time, and cost [23,24]. The nerve block techniques studied in our trial are simpler, quicker to perform, and less invasive, making them more accessible for widespread clinical use, especially in centers without advanced thoracoscopic equipment. In addition, the analgesic effect of INC is not immediate; axonal degeneration may take 12 to 24 h to complete, during which patients may still experience significant



**Fig. 2.** Comparison of numerical rating scale (NRS) pain scores over time at rest and during movement between SAPB and TPVB after extubation. The data are expressed as the median (interquartile range). Group T: thoracic paravertebral block group; group S: serratus anterior plane block group;

postoperative pain [23,25]. As part of bridging analgesia during this latency period, it is reasonable to choose nerve block techniques that are simpler to perform, less invasive, and associated with higher safety [26]. Future research could explore combining cryoanalgesia with regional nerve blocks to establish optimized, multimodal analgesia protocols for acute postoperative pain [26]. The findings of our study may serve as a useful reference for designing such comprehensive strategies.

Although there is no single best method, multimodal analgesia should be implemented without requiring advanced surgical or anesthetic techniques while achieving excellent analgesic outcomes. Multimodal analgesia pathways often include regional anesthesia. As a

decade-old technique, ultrasound-guided TPVB has a relatively long learning curve and can result in pneumothoraxes and nerve root injuries, particularly when performed by novices. In contrast, SAPB allows analgesic effects to be achieved through relatively simple procedures at a safe distance from the spinal cord and pleura. The puncture target for SAPB is shallow enough to minimize the risk of damage to major blood vessels and nerves, which makes it a safer alternative to TPVB and reduces the possibility of complications. Notably, all blocks in our experiment were performed by an attending anesthesiologist with extensive experience in both techniques in pediatric patients. In experienced hands, ultrasound-guided paravertebral nerve blocks appear to



**Table 2**

Comparison of outcomes between the groups. Values are mean  $\pm$  SD, median (IQR [range]) or number (proportion).

|                                                                                        | SAPB group<br>N = 37               | TPVB group<br>N = 37               | Effect<br>size | 95 % (CI)                                                         |
|----------------------------------------------------------------------------------------|------------------------------------|------------------------------------|----------------|-------------------------------------------------------------------|
| Comparative efficacy of SAPB versus TPVB on primary outcomes: tests for noninferiority | 1 (1,2)                            | 1 (0.5,2)                          | 0.027          | (−0.42, 0.47)                                                     |
| <b>Secondary outcomes</b>                                                              |                                    |                                    | <b>P value</b> | <b>Risk difference, odds ratio or median difference (95 % CI)</b> |
| Intraoperative sufentanil, $\mu\text{g}\cdot\text{kg}^{-1}$                            | 0.54 $\pm$ 0.96                    | 0.48 $\pm$ 0.10                    | 0.005*         | 0.07(0.02, 0.11)                                                  |
| Total morphine-equivalent doses postoperative 24 h $\text{mg}\cdot\text{kg}^{-1}$      | 1.12<br>(1.06,1.21)<br>[0.97,1.33] | 1.11<br>(1.07,1.20)<br>[0.96,1.48] | 0.863          | 0(−0.05,0.05)                                                     |
| Time to first rescue analgesia, (min)                                                  | 74<br>(42,3173.5)<br>[26,1841]     | 84<br>(62,5155.5)<br>[25,1509]     | 0.559          | −8.5(−35,24)                                                      |
| PCA presses number, (effective, n)                                                     | 26(11.5, 39.25)<br>[5,95]          | 27(17, 41.5)<br>[0,64]             | 0.515          | −3<br>(−10.00,6.00)                                               |
| Hypotension, n (%)                                                                     | 3(8.1)                             | 11(29.7)                           | 0.018*         | 4.80<br>(1.21,18.96)                                              |
| Urinary retention, n (%)                                                               | 4(10.8)                            | 0(0)                               | 0.123          | 0.89(0.80,1.00)                                                   |
| Nausea and vomiting, n (%)                                                             | 9(24.3)                            | 8(21.6)                            | 0.782          | 0.86(0.29,2.54)                                                   |
| Use of oral analgesics, n (%)                                                          | 7(18.9)                            | 5(13.5)                            | 0.528          | 0.67(0.19,2.34)                                                   |
| Local anesthetic toxicity                                                              | 0                                  | 0                                  | /              |                                                                   |

PCA: Patient Controlled Analgesia; TPVB: thoracic paravertebral block; SAPB: serratus anterior plane block; CI: confidence interval.

\* Compared to control group,  $P < 0.05$ .

present minimal risk [27], but both theoretical considerations and limited evidence suggest that SAPB is easier to learn and administer. [14,28–30] In clinical practice, practitioners with less experience may prefer to perform less invasive fascial plane blocks because of risk/benefit ratio concerns.

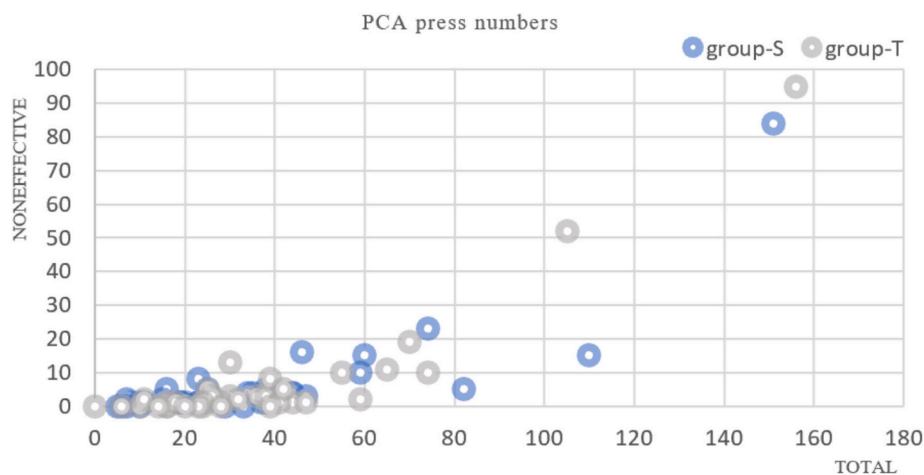
The evidence on the analgesic efficacy of SAPB versus TPVB is still

conflicting. The targets of both blocks are the same (i.e., the thoracic and intercostal nerves). TPVB is assumed to be more reliable in targeting nerve roots by blocking spinal nerves in the paravertebral space [27]. In contrast, SAPB relies on the spread of local anesthetics to intercostal nerves travelling within the fascial plane [31]. In patients undergoing chest wall surgery, it is reported that both blocks generally offer comparable analgesic efficacy [14,30]. In the pediatric population, a study comparing TPVB and SAPB for children undergoing video-assisted thoracoscopic surgery revealed that SAPB provided effective post-operative pain management similar to that of TPVB but was technically easier and faster to perform, making it a safe alternative to TPVB [29]. Our study is the first randomized clinical trial to compare the effectiveness of SAPB and TPVB in this surgical population and our results align with these findings.

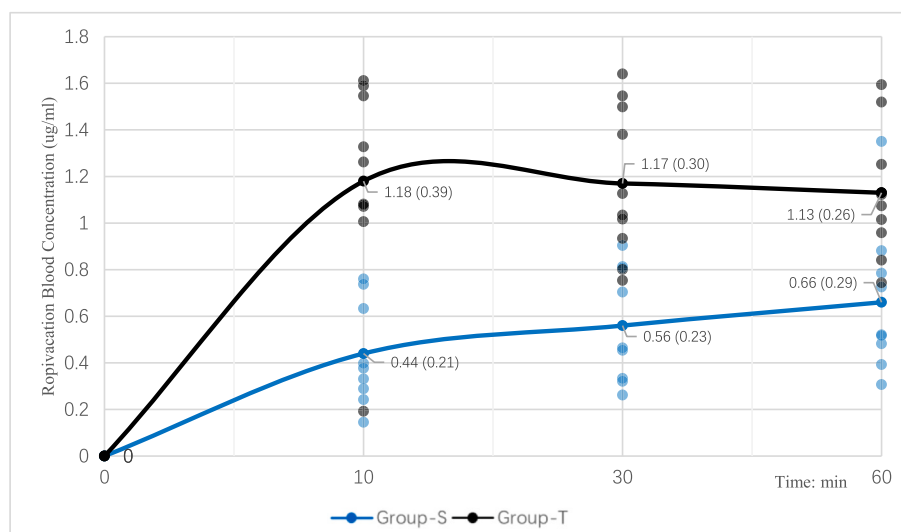
Similar to previous reports [29,32], paravertebral block preparation took longer than SAPB preparation did (median: 13 vs. 6 min). This seven-minute difference is likely not clinically significant. However, SAPB can be administered during intubation in the supine position, reducing risks linked to repositioning, such as dislodgement of intubation tubes or invasive arterial catheters, as well as fractures. Additionally, this time efficiency may increase surgeon satisfaction and improve surgical turnover rates.

Our previous study first reported arterial plasma ropivacaine concentrations after SAPB [33]. We monitored the plasma concentration dynamics of ropivacaine in children undergoing congenital heart surgery with deep SAPB using  $2.5 \text{ mg kg}^{-1}$  0.2 % ropivacaine, and found a mean peak concentration ( $C_{\text{max}}$ ) of  $2.1 \mu\text{g mL}^{-1}$  in arterial blood at 10 min postblock. For VATS patients, Tang et al. reported an average arterial  $C_{\text{max}}$  of  $1.25 \mu\text{g mL}^{-1}$  at 0.9 h postblock with 0.25 % ropivacaine [34]. In this study, adding epinephrine (1:200,000) to ropivacaine lowered the arterial  $C_{\text{max}}$  and delayed the time to peak concentration ( $T_{\text{max}}$ ) after SAPB. These results indicate that SAPB may prevent a rapid increase in the plasma ropivacaine concentration compared with TPVB. However, significant interindividual variability in peak arterial plasma levels was observed in both groups. Given the small sample size, further research is needed to clarify the pharmacokinetic profile of ropivacaine, especially with epinephrine, for interfascial blocks.

Our study has several limitations. First, as all the blocks were performed under anesthesia and reliable sensory assessment data in pediatric patients presents practical challenges, we did not systematically evaluate the onset and duration of the sensory block, such as the duration of chest wall numbness. Additionally, pain assessment was the sole primary outcome used to evaluate non-inferiority. Future studies should consider including opioid administration as a dual primary outcome to broaden the applicability of the results. Furthermore, plasma concentrations were measured only at 10, 30, and 60 min after blocks, meaning



**Fig. 3.** Individual PCA press numbers in the SAPB and TPVB groups within 72 h.



**Fig. 4.** Individual ropivacaine plasma concentrations in the SAPB and TPVB (B) groups up to 60 min after the injection of  $2.5 \text{ mg kg}^{-1}$  ropivacaine 0.25 %. The data are expressed as the means (standard deviations).

Group T: thoracic paravertebral block group; group S: serratus anterior plane block group;

that safety was not assessed throughout the entire study period.

In conclusion, SAPB provides non-inferior analgesia to TPVB in children undergoing Nuss surgery, with comparable pain scores at rest and during coughing 24 h postoperatively. However, SAPB was associated with increased intraoperative opioid consumption. In clinical practice, the choice between these two blocks involves considering multiple factors, such as procedural time, the risk of complications, and patient-specific considerations. As with any medical intervention, the benefits must be weighed against the potential risks to determine the most appropriate approach for each patient.

#### CRediT authorship contribution statement

**Yi He:** Writing – review & editing, Writing – original draft, Project administration, Conceptualization. **Mingzhe Xu:** Methodology, Investigation, Formal analysis, Data curation. **Zhi Li:** Methodology, Investigation. **Jiaqi Yu:** Methodology, Investigation. **Qian Li:** Supervision, Project administration. **Yunxia Zuo:** Supervision, Project administration. **Yi Kang:** Methodology. **Bin Du:** Writing – review & editing, Supervision, Project administration, Conceptualization.

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#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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